

regarding the specific skills and contexts required for AI integration. Additionally, the GENOE index considers the complex interplay of tasks within occupations and integrating the ethical and social contexts organically.

Another contribution of our paper is the novel use of large language models to generate synthetic assessments that aim to replicate and extend expert assessments. Many studies rely on expert surveys or interviews to build analytical tools like indexes, including in the labor automation area (Frey and Osborne, 2017; Felten *et al.*, 2021; Tolan *et al.*, 2021). Synthetic AI surveys offer several advantages. They can efficiently process and synthesize vast amounts of information from diverse sources very rapidly, enabling a larger volume of questions to be addressed in a shorter time frame. Additionally, synthetic AI surveys can be easily scaled to include extensive sample sizes or repeated iterations, allowing greater granularity of the collected data.

While recent research has begun to explore the use of AI in workforce analysis, our methodology offers a novel approach in this evolving field. For example, Eloundou *et al.* (2024) combined human expert insights with AI-generated assessments to evaluate AI’s potential for enhancing task efficiency across occupations. Our work is a step forward in this approach by exclusively and systematically using ChatGPT to build the entire index through multiple iterations and varying model parameters.

To validate the accuracy of this methodology, we replicated the well-known exposure index of Felten *et al.* (2021), replacing mTurk surveys with synthetic AI surveys. Our approach successfully replicated Felten’s AI Occupational Exposure Index (AIOE) with a correlation of 0.99, suggesting our methodology is robust and reliable for these types of tasks and providing evidence of another area where AI-driven automation is highly feasible.

The remainder of this paper is organized as follows. After this introduction, Section 2 details our data sources and methodology used to build the GENOE index. Section 3 presents the GENOE index, including its key findings and implications. This section also includes several robustness checks to validate our results and replicates other human-like expert assessment using our novel synthetic survey methodology. Section 4 explores how the index can be used in practical applications, analyzing occupational exposure in the U.S. and Mexico labor market. Finally, Section 5 discusses some considerations and limitations of our research, while Section 6 concludes with implications and directions for future research.

2 DATA AND METHODOLOGY

We draw data from the O*NET database, which contains detailed descriptions and attributes for 1,016 occupations in the U.S. labor market under the O*NET-SOC 2019 classification. Occupational characterizations are built and continuously updated from surveys and input from multiple sources, including job analysts, incumbent workers, and industry experts. O*NET attributes—such as abilities, skills, knowledge, and tasks—are divided into six major domains: Worker Characteristics, Worker Requirements, Experience Requirements, Occupational Requirements, Work-

veys. To ensure compatibility with standard occupational classifications, we translated these estimations to the SOC-18 classification using O*NET’s crosswalks for a total of 759 occupations.

To enhance our understanding of the GENOE’s effectiveness and to validate our methodology, we conducted several additional exercises. Firstly, we measured task-level exposure by assessing the proportion of tasks at risk within each occupation, allowing us to identify roles where AI might significantly impact certain tasks without necessarily replacing the entire job. We also evaluated tasks in isolation and compared the results with our holistic occupation-level approach to understand the value of considering tasks within their broader occupational context. Additionally, we tested the effect of withholding the occupation name from the model to gauge the importance of occupational context in the AI’s assessment. To incorporate broader societal considerations, we explicitly prompted the model to consider ethical and social factors in its assessments.

Furthermore, we conducted follow-up surveys, asking the AI to explain its reasoning behind the exposure estimates, effectively “opening the black box” of its decision-making process. These additional analyses serve multiple purposes: they validate our methodology by providing alternative perspectives on AI exposure, offer insights into the factors influencing AI’s potential impact on various occupations, and help us understand the decision-making process of the AI model. This approach not only strengthens the robustness of our GENOE index but also potentially reveals any biases or limitations in the AI’s assessments, providing a more reliable picture of occupational exposure to AI.

To further validate our approach, we applied our synthetic survey methodology to replicate the Felten *et al.* (2021) index of occupational exposure. In this exercise, we substituted the original human expert surveys with synthetic surveys conducted using ChatGPT4o. The primary objective was to assess the capability of our AI-driven methodology to reproduce results comparable to those obtained through traditional human expert assessments. This replication serves as a crucial test of our approach’s reliability and accuracy, offering insights into the potential of large language models to emulate human expertise in evaluating occupational exposure to AI.

3 RESULTS

3.1 Baseline Estimations

Figure 1 presents a comprehensive view of AI exposure across 759 occupations, illustrating the average, median, 25th percentile, and 75th percentile for the three defined time horizons. The analysis reveals a progressive increase in average AI exposure: 0.28 within one year, rising to 0.38 over five years, and reaching 0.44 over a decade. This pattern suggests that as AI technologies advance and mature, their potential to reshape various professions grows significantly.⁵

⁵Complete descriptive statistics can be found in Appendix Table A.1.

Extrapolating these findings to Latin America and the Caribbean, we estimate that approximately 84 million jobs are exposed to AI within one year, rising to 114 million in 5 years, and 132 million in 10 years. Extending this average exposure to global labor markets suggests that approximately 980 million, 1.33 billion, and 1.54 billion jobs may be exposed to AI over the one-, five-, and 10-year horizons, respectively. However, it is important to note that this projection assumes uniform AI adoption rates and labor market structures across countries, which may not accurately reflect the global economic landscape.

5 LIMITATIONS AND CONSIDERATIONS

While our methodology offers a novel and comprehensive approach to assessing occupational exposure to AI, it is important to acknowledge its limitations. Firstly, the assessments are based on the knowledge and potential biases inherent in the GPT-4o model. Although state-of-the-art, this model may have limitations in understanding certain specialized fields or future AI developments. Additionally, predictions about future AI capabilities, especially in the five and 10-year horizons, are inherently uncertain and may not account for unforeseen breakthroughs or obstacles in AI development. Our method attempts to capture broader contextual factors, but it may not fully account for all social, economic, and regulatory factors that could influence the adoption of AI in various occupations. Lastly, while the O*NET task descriptions are detailed, they may not capture all ways in which tasks are performed in real-world settings, potentially affecting the accuracy of AI exposure estimates.

Another limitation of our index—and indeed, of all indexes that aim to capture occupational exposure to technology—is that it does not account for general equilibrium effects that can influence social preferences and the demand for certain jobs. For instance, while firefighters currently exhibit low exposure estimates to technological displacement, advances in electrical systems, domestic safety features, and industrial processes could lead to a significant reduction in the occurrence of fires. This, in turn, could decrease the overall demand for firefighters, despite their low exposure to direct technological replacement. Thus, our index may not fully capture how broader technological progress can indirectly impact the demand for certain occupations through changes in societal needs and preferences.

Despite these limitations, the GENOE index offers a unique and holistic perspective on the future of work in the age of artificial intelligence. By combining a robust data source, a novel AI-driven survey methodology, and comprehensive validation exercises, we have created a valuable tool for researchers, policymakers, and industry leaders to anticipate and prepare for the potential impacts of AI on the labor market. The index’s strengths lie in its comprehensive coverage, consistent evaluation criteria, and ability to capture more complex aspects like interactions between tasks and contextual factors. As such, while acknowledging its constraints, we believe the GENOE index provides a significant contribution to our understanding of occupational exposure to AI and can inform strategic decision-making in workforce planning and policy development.

6 CONCLUDING REMARKS

This paper introduces the AI-Generated Index of Occupational Exposure (GENOE), a novel measure of susceptibility to job replacement by artificial intelligence. Using synthetic AI surveys conducted with advanced language models, we estimate the likelihood of job replacement by AI over one, five, and 10-year horizons. Our findings reveal a significant increase in exposure over time, with the average likelihood of job replacement rising from 0.28 in the next year to 0.38 and 0.44 over the next five and 10 years, respectively, underscoring the accelerating impact of AI on the labor market.

The GENOE methodology of synthetic AI surveys leverages the extensive knowledge and analytical capabilities of AI models to provide expert-like assessments. This approach offers a comprehensive view of occupational exposure by considering not only the tasks performed within each occupation but also ethical and social contexts that influence the likelihood of AI-driven automation. By doing so, it overcomes the limitations of previous measures by incorporating a more holistic task-based analysis and integrating other occupation-specific, social, and ethical considerations more organically and with fewer assumptions. This contrasts with prior methods that isolated tasks or abilities separately or relied on more predefined criteria for measuring exposure.

We validated the consistency of the index by varying key elements of the methodology and the prompts, such as calculating the task-by-task exposure, occupation name omission, and explicitly nudging the model to consider ethical and social resistance to automation considerations. Additionally, follow-up surveys with AI models uncovered key patterns including understanding and projections of current and expected technological capabilities, the routine versus non-routine nature of tasks, ethical considerations, regulatory environments, and cost-benefit analyses.

The GENOE index offers significant and practical uses and implications for academia, policymakers, industries, and individual workers. By identifying which types of tasks, occupations, and industries are most susceptible to AI-driven automation, the GENOE index can inform strategic decisions and policy formulations. For academia, it offers a robust framework for future research on labor market dynamics and technology's impact, opening new avenues for interdisciplinary studies on AI and employment.

Policymakers can use the index to develop targeted interventions and support measures for workers in highly exposed occupations, ensuring a smoother transition in the face of technological change. This data-driven approach can inform education and training policies, unemployment insurance programs, and economic development strategies. Industries can leverage the index to guide strategic decision on workforce development, technology integration, and long-term business planning. It can help companies anticipate skill gaps, plan for reskilling initiatives, and optimize human-AI collaboration.

For individual workers, the index provides valuable insights for career planning and skill development, helping them prepare for potential shifts in their occupational landscape. In the